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MULTIPOINT INTERPOLATED DFT METHOD FOR FREQUENCY ESTIMATION

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ABSTRACT

The accuracy of the frequency estimation of a multifrequency signal component by Interpolated Discrete Fourier Transform (IpDFT) method is affected by systematic errors. In [1] a Weighted Multipoint Interpolated Discrete Fourier Transform (WMIpDFT) method has been proposed in order to reduce these errors. This method uses only the rectangular and the 2-term maximum sidelobe decay windows. In this paper the results of the method proposed in [1] are generalized for the maximum sidelobe decay windows with higher order than two, which are used to improve the performance by reducing the systematic errors. When the H-term maximum sidelobe decay window ($H \geq 2$) is used and the number of interpolation points is $(2J + 1)$, with $1 \leq J \leq H-1$, the analytical formula to estimate the frequency is derived. Then, the performance of the obtained WMIpDFT method is analyzed by means of computer simulations.

Index Terms— Discrete Fourier transform (DFT), interpolation, parameter estimation, signal processing

1. INTRODUCTION

One of the most important parameter of a multifrequency signal component is their frequency values. For this reason it is very useful to estimate this parameter with a very high accuracy. Methods used for this purpose can be classified either in time-domain (parametric) or in frequency-domain (nonparametric) methods [2]. The frequency-domain methods do not require a rigid signal model and can be easily implemented due to the availability of Fourier transform packages. From these reasons in many engineering applications the frequency-domain methods are used to estimate the parameters of a multifrequency signal component.

One of the frequency-domain methods used to estimate the frequency values of a multifrequency signal component in noncoherent sampling mode is the Interpolated Discrete Fourier Transform (IpDFT) method [3]–[6]. This method provides very accurate frequency estimates and its best performances are obtained using the maximum sidelobe decay windows (or the class I Rife-Vincent windows) [7], [8] since in this case the frequency is estimated by means of analytical formulas [5], [6].

The accuracy of the frequency estimation by IpDFT method is affected by the systematic errors due to the contributions from other components in multifrequency signal. In order to reduce these errors, two Weighted Multipoint IpDFT (WMIpDFT) methods have been proposed in [1] and [9]. The WMIpDFT method described in [1] uses only the rectangular window and the 2-term maximum sidelobe decay window (or Hann window) whereas the WMIpDFT method proposed in [9] uses the maximum sidelobe decay windows with an order, H, higher than or equal to two.

This paper focuses on the generalization of the WMIpDFT method proposed in [1] and using the maximum sidelobe decay windows with higher order ($H \geq 2$). Then, the performance of the obtained WMIpDFT method is compared by means of computer simulations with the one of the WMIpDFT method proposed in [9]. Two cases are considered: multifrequency signal without noise and with a white Gaussian noise.

2. THEORETICAL BACKGROUND

Let us consider a multifrequency signal sampled at a known frequency, f_s, i.e.,

$$x(m) = A_0 + \sum_{k=1}^{K} A_k \sin\left(2\pi \frac{f_k}{f_s} m + \varphi_k\right), \qquad m = 0,1,2,\ldots,M-1 \tag{1}$$

where K is the number of frequency components; A_k, f_k, and φ_k are the amplitude, frequency, and phase of the kth component respectively; A_0 is the offset and M is the acquisition length. To satisfy the Nyquist criterion, frequencies f_k, k = 1, 2,..., K, are smaller than $f_s/2$. The relationship between the frequencies f_k and f_s is given by

$$\frac{f_k}{f_s} = \frac{\lambda_k}{M} = \frac{l_k + \delta_k}{M}, \quad k = 1, 2, \ldots, K \tag{2}$$

in which λ_k is the number of recorded sine-wave cycles associated with the kth component, l_k and δ_k are, respectively, the integer and the fractional parts of λ_k. δ_k is related to the noncoherent sampling mode with $-0.5 \leq \delta_k < 0.5$.

For $\delta_k = 0$, the sampling process is coherent with the sine-wave associated with the kth component (coherent sampling mode) and (2) represents the coherent sampling relationship between f_k and f_s.

For $\delta_k \neq 0$ the sampling process is noncoherent with the sine-wave associated with the kth component (noncoherent sampling mode), and the coherent sampling relationship is not fulfilled. To reduce the spectral leakage errors which arises in the noncoherent sampling mode, windowing approach is used, leading to the spectral analysis of $x_w(m) = x(m)\cdot w(m)$, where $w(m)$ is the window sequence. The Discrete-Time Fourier Transform (DTFT) of $x_w(m)$ is given by

$$\begin{aligned} X_w(\lambda) &= A_0 W(\lambda) \\ &+ \sum_{k=1}^{K} \frac{A_k}{2j}\left[W(\lambda - \lambda_k)e^{j\varphi_k} - W(\lambda + \lambda_k)e^{-j\varphi_k}\right] \\ &\qquad \lambda \in [0, M) \end{aligned} \tag{3}$$

where λ is the continuous normalized frequency expressed in frequency bins and $W(\lambda)$ is the DTFT of $w(m)$.The second term in the square brackets of (3) represents the image part of the spectrum.

The spectral component magnitudes centered on the frequency f_k is a sum of two terms. The first term is due to the short-range spectrum leakage contribution of the window spectrum weighted by $A_k/2$ and the second term is the term $|\Delta(l)|$ due to the long-range leakage of the image part of the spectrum associated with the kth component and to the long-range leakage contributions from other components in the multifrequency signal [10]. The terms $|\Delta(l)|$ decrease as the frequency increases. In addition, if they have a sine function in the kernel they change sign at successive magnitudes. This property is valid when using the maximum sidelobe decay windows. The H-term maximum sidelobe decay window ($H \geq 2$) has the most rapidly decaying sidelobes, which is equal to 6(2H-1) dB/octave, from all the H-term cosine windows [7], [8]. This window is defined by

$$w(m) = \sum_{h=0}^{H-1} (-1)^h a_h \cos\left(2\pi h \frac{m}{M}\right), \quad m = 0, 1, \ldots, M-1 \tag{4}$$

where a_h are the window coefficients given by [8]

$$a_0 = \frac{C_{2H-2}^{H-1}}{2^{2H-2}}, \quad a_h = \frac{C_{2H-2}^{H-h-1}}{2^{2H-3}}, \quad h = 1, 2, \ldots, H-1 \tag{5}$$

where $C_m^p = \dfrac{m!}{(m-p)!\,p!}$.

For M >> 1, $W(\lambda)$ can be approximated with very high accuracy by [6]

$$W(\lambda) = \frac{M \sin(\pi\lambda)}{2^{2H-2}\pi} e^{-j\frac{\pi(M-1)}{M}\lambda} \frac{(2H-2)!}{\lambda \prod_{h=1}^{H-1}\left(h^2 - \lambda^2\right)}. \tag{6}$$

When a maximum sidelobe decay window is used, it has been shown that $|X_w(l_k - 1)|$, $|X_w(l_k)|$ and $|X_w(l_k + 1)|$ are given by [10]

$$\begin{aligned} |X_w(l_k - 1)| &= \frac{A_k}{2}|W(-1-\delta_k)| \mp |\Delta(l_k - 1)| \\ |X_w(l_k)| &= \frac{A_k}{2}|W(-\delta_k)| \pm |\Delta(l_k)| \\ |X_w(l_k + 1)| &= \frac{A_k}{2}|W(1-\delta_k)| \mp |\Delta(l_k + 1)|. \end{aligned} \tag{7}$$

Moreover, for large l_k the values $|\Delta(l_k \pm i)|$, i = 0, 1,..., H are very close.

3. PROPOSED WMIpDFT METHOD

Let us consider the (2J + 1) - point interpolation with all the magnitudes $|X_w(l_k \pm i)|$, i = 0, 1, 2,..., J, involved in the interpolation localized within the spectrum main lobe of the kth signal frequency. The width of the spectrum main lobe of the kth signal frequency expressed in bin is equal to 2H. Thus, all the interpolation points are within the spectrum main lobe of the kth signal frequency if $J \leq H-1$.

From the observations associated with the $|\Delta(l)|$ sign, it follows that summing the magnitudes $|X_w(l_k \pm i)|$, i = 0, 1,..., J, the long-range leakage tails are subtracted and so, theirs influences are reduced.

Let us define the ratio α_k as follows

$$\begin{aligned} \alpha_k = &\frac{\sum_{i=1}^{J} C_{2J-2}^{J-i}\left[|X_w(l_k+i)| - |X_w(l_k-i)|\right]}{\sum_{i=1}^{J} C_{2J}^{J-i}\left[|X_w(l_k-i)| + |X_w(l_k+i)|\right] + C_{2J}^{J}|X_w(l_k)|} \\ &- \frac{\sum_{i=1}^{J-2} C_{2J-2}^{J-i-2}\left[|X_w(l_k+i)| - |X_w(l_k-i)|\right]}{\sum_{i=1}^{J} C_{2J}^{J-i}\left[|X_w(l_k-i)| + |X_w(l_k+i)|\right] + C_{2J}^{J}|X_w(l_k)|}. \end{aligned} \tag{8}$$

Regarding the denominator, it should be noted that the weights associated with the magnitudes $|X_w(l_k \pm i)|$, i = 0, 1,..., J are obtained using the 2J – order finite difference of $|\Delta(l_k - J)|$, $|\Delta^{2J}(l_k - J)|$. The first order finite differences of $|\Delta(l_k \pm i)|$, $i = 0, 1, \ldots, J$, are given by $|\Delta^1(l_k \pm i)| = |\Delta(l_k \pm i + 1)| - |\Delta(l_k \pm i)|$, $i = 0, 1, \ldots, J$. The p–order finite differences of $|\Delta(l_k \pm i)|$, $i = 0,1,\ldots, J$, are given by $|\Delta^p(l_k \pm i)| = |\Delta^{p-1}(l_k \pm i + 1)| - |\Delta^{p-1}(l_k \pm i)|$, $p = 2, 3, \ldots, 2J$. $|\Delta^{2J}(l_k - J)|$ is given by [11]

$$\left|\Delta^{2J}(l_k - J)\right| = \sum_{i=1}^{J}(-1)^{J-i} C_{2J}^{J-i}\left[\left|\Delta(l_k - i)\right| + \left|\Delta(l_k + i)\right|\right] + (-1)^J C_{2J}^{J}\left|\Delta(l_k)\right|. \tag{9}$$

Using the equality $C_{2J-2}^{J-i} + C_{2J-2}^{J-i-2} = C_{2J}^{J-i} - 2C_{2J-2}^{J-i-1}$, regarding the numerator, it follows that the weights associated with the magnitudes $|X_w(l_k \pm i)|$, $i = 0, 1, \ldots, J$, are obtained using $|\Delta^{2J}(l_k - J)|$ and the $(2J - 2)$-order finite difference of $|\Delta(l_k - J + 1)|$, $|\Delta^{2J-2}(l_k - J + 1)|$.

Since the influence of the long-range leakage tails is neglected from (3) α_k can be approximated by

$$\alpha_k \cong \frac{\sum_{i=1}^{J} C_{2J-2}^{J-i}\left[\left|W(i-\delta_k)\right| - \left|W(i+\delta_k)\right|\right]}{\sum_{i=1}^{J} C_{2J}^{J-i}\left[\left|W(i-\delta_k)\right| + \left|W(i+\delta_k)\right|\right] + C_{2J}^{J}\left|W(\delta_k)\right|} - \frac{\sum_{i=1}^{J-2} C_{2J-2}^{J-i-2}\left[\left|W(i-\delta_k)\right| - \left|W(i+\delta_k)\right|\right]}{\sum_{i=1}^{J} C_{2J}^{J-i}\left[\left|W(i-\delta_k)\right| + \left|W(i+\delta_k)\right|\right] + C_{2J}^{J}\left|W(\delta_k)\right|}. \tag{10}$$

Using (6) and after some algebra, the right term of the above expression becomes equal to (see (A.9) from Appendix)

$$\alpha_k \cong \frac{\delta_k}{H + J - 1}. \tag{11}$$

From the above expression it follows that δ_k can be estimated by

$$_{2J+1}\hat{\delta}_k = (H + J - 1)\alpha_k. \tag{12}$$

l_k is the index value that corresponds to the maximum of $|X_w(\lambda)|$, for integer values (i.e. for $\lambda = 1,2, \ldots, M/2 - 1$) associated with the kth component. So, it can be very easily determinate by means of a maximum search routine if the sine-wave frequency signal-to-noise ratio is above threshold [4]. l_k is so exactly known as well as M and f_s. Thus, from (2) it is obvious that the frequency estimation accuracy is linked to the accuracy of δ estimation obtained by (12).

It should be noted that by the proposed method δ_k is estimated by a simple analytical formulas, in comparison with the WMIpDFT method proposed in [9], where for 3, 5 and 7 interpolation points δ_k is estimated by analytical formulas, which for 5 and 7 interpolation points are complicated. Another important observation is the fact that for 3-point interpolation the formula for estimating δ_k is the same in both WMIpDFT methods.

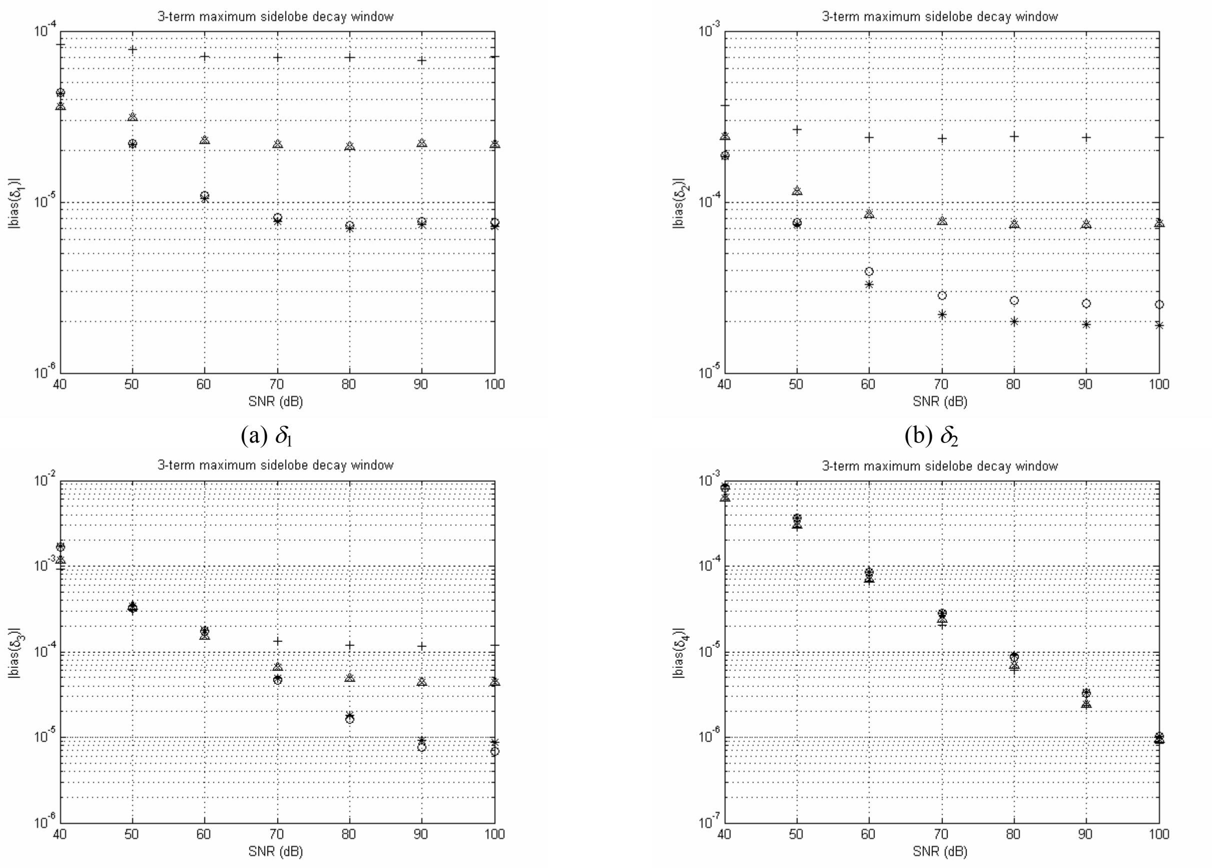
4. COMPUTER SIMULATION

The aim of this section is to compare by means of computer simulations the performance of the proposed WMIpDFT method with the one of the WMIpDFT method proposed in [9]. For this purpose a modeled multifrequency signal is used [9]. This is characterized by: $A_0 = 0.1$, $A_1 = 2$, $A_2 = 0.5$, $A_3 = 0.07$, $A_4 = 0.1$, $l_1 = 5$, $l_2 = 13$, $l_3 = 24$, $l_4 = 125$ and $M = 1024$. First the signal is not affected by noise. Are considered $\delta_1 = \delta_2 = \delta_3 = \delta_4 = \delta$, where δ varies in the range $[-0.5, 0.5)$ by a fixed increment and the phases φ_1, φ_2, φ_3 and φ_4 are uniformly distributed in the range $[0, 2\pi)$ rad. For each value of δ the maximum of the modulus of the absolute error of δ_k, $|\Delta\delta_k|_{\max}$ occurring during the variation of phases is retained. It has been shown that the proposed WMIpDFT method reduces the systematic errors as the number of points involved in the interpolation increases (i.e. J increases). Nevertheless, it was proven by simulation that the systematic errors decrease as H increases. In addition, the accuracy of the δ_k estimates depends on the mutual frequency component span [9]. To avoid spectral interferences from the nearby components it is necessary to fulfill this relationship: $(l_{k+1} - l_k) > 2H + 1$, $k = 0, 1, \ldots, K$ (in which l_0 corresponds to the DC component) [9].

Due to its real-life importance a white Gaussian noise with zero mean and variances σ_n^2 is added to the multifrequency signal. Taking into account this noise, the standard deviation of the δ_k estimates obtained by the IpDFT method with H-term maximum sidelobe decay window is approximated by [6]

$$\sigma_{\delta_k} = \frac{2\pi\delta_k}{A\sin(\pi\delta_k)}\frac{H - |\delta_k|}{(2H-1)!}\prod_{h=1}^{H-1}\left(h^2 - \delta_k^2\right) \times \sqrt{\frac{C_{4H-4}^{2H-2}}{2M}}\sqrt{\frac{2(4H-3)\left(\delta_k^2 - |\delta_k|\right) + 2H^2 - 1}{2H-1}}\,\sigma_n. \tag{13}$$

To the simulated multifrequency signal an additive white Gaussian noise with zero mean and σ_n standard deviation is added. σ_n is established as a function of the Signal-to-Noise Ratio (SNR). SNR varies in the range [40, 100] dB with a step of 10 dB. The values of δ_k are set as follows [9]: $\delta_1 = 0.1$, $\delta_2 = -0.4$, $\delta_3 = 0.3$, and $\delta_4 = -0.25$. For each SNR, 100 value sets for the phases φ_1, φ_2, φ_3, and φ_4 uniformly distributed in the range $[0, 2\pi)$ rad are used. For each set, 1000 runs are performed to calculate the mean value of δ_k and the worst bias of δ_k is, finally retained. The 3-term maximum sidelobe decay window is employed. Fig. 1 shows the modulus of the bias of δ_k estimates obtained by the WMIpDFT methods and by the IpDFT method [6] as a function of SNR. In the WMIpDFT methods the 3 and 5-point interpolation are used.



(a) δ_1 (b) δ_2 (c) δ_3 (d) δ_4

Figure 1. Modulus of the bias of the δ_k estimates as a function of SNR when the 3-term maximum sidelobe decay window is used. δ_k is estimated by the proposed WMIpDFT method ('o'-5-point interpolation, 'Δ'-3-point interpolation), by the WMIpDFT method proposed in [9] ('*'-5-point interpolation, 'x'-3-point interpolation) and by the IpDFT method ('+').

In Fig. 1, it can be observed that the results obtained for 3-point interpolation are the same since the estimating formula is the same and for 5-point interpolation the results obtained by both WMIpDFT methods are very close. Thus, it follows that the effectiveness of both WMIpDFT methods in reducing the systematic error is practically the same. When the WMIpDFT methods are used, for a given value of J the accuracy of δ_k estimates depends on the position of the frequency component (i.e. on l_k) and on SNR. In addition, for a given J the accuracy of the δ_k estimates depends, as for a signal without noise, on the mutual frequency component span.

For large l_k, the systematic errors become smaller than the errors due to the noise and so, the accuracy of δk estimates is practically unchanged (see Fig. 1d). This behavior is also obtained for small SNR (smaller than 40 dB). For this reason the WMIpDFT methods are well suited to be use when lk is not so large and for relative greater SNR (i.e. for signal corrupted with relative small power noise). In these cases high accurate estimates are obtained.

Moreover, the efficiencies of the proposed WMIpDFT method and of the WMIpDFT method proposed in [9], evaluated by means of the ratio between the variance of the δ_k estimates and the corresponding unbiased Cramer-Rao (CR) lower bound, have been compared.

Under the assumption of a white Gaussian noise, the unbiased CR lower bound for the δ_k estimates is [4]

$$\mathrm{var}\left(\hat{\delta}_k\right)_{CR} \cong \frac{6}{\pi^2}\frac{\sigma_n^2}{MA_k^2}. \tag{14}$$

Fig. 2 shows the variance of δ_3 estimates obtained by the WMIpDFT methods and by the IpDFT method normalized to the corresponding CR lower bound as a function of δ_3. Fig. 2 also presents the theoretical variance of δ_3 estimates obtained by the IpDFT method [see (13)] normalized to the corresponding CR lower bound as a function of δ_3. SNR is set to 70 dB. δ_3 varies in the range [0, 1) with a step of 1/25. The phases of the components are set as follows: $\varphi_1 = 0$ rad, $\varphi_2 = \pi/3$ rad, $\varphi_3 = 3\pi/4$ rad and $\varphi_4 = \pi/6$ rad. For each δ_3, 5000 runs are done. All other parameters are set as in Fig. 1.

In Fig. 2, it can be observed that when the WMIpDFT methods are used the variance of δ_3 estimates normalized to the corresponding CR lower bound increases as the number of interpolation points increases. In addition, it can be observed that for the 5-point interpolation the proposed WMIpDFT method is somewhat more efficient than the method WMIpDFT proposed in [9].

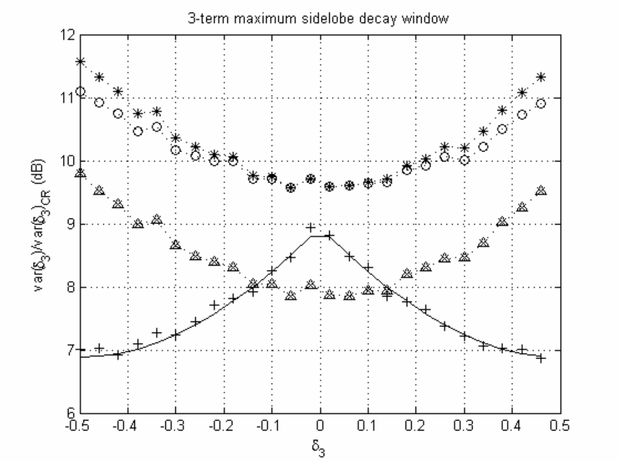


Figure 2. Ratio between the variance of the δ_3 estimates obtained by the proposed WMIpDFT method ('o'-5-point interpolation, 'Δ'-3-point interpolation), by the WMIpDFT method proposed in [9] ('*'-5-point interpolation, 'x'-3-point interpolation) and by the IpDFT method ('+') normalized to the corresponding CR lower bound the as a function of δ_3. Continuous line represents the theoretical ratio between the variance of the δ_3 estimates obtained by the IpDFT method and the corresponding CR lower bound as a function of δ_3.

This behavior was obtained also for the δ_1, δ_2 and δ_4 estimates. Simulations have been carried out for different values of phases φ_k (k = 1, 2, 3, 4), amplitudes A_k (k = 1, 2, 3, 4) and SNR between 40 and 100 dB and a similar behavior as in Fig. 2 is obtained. If the number of interpolation increases, then the noise increases and thus, the statistical efficiency of the WMIpDFT methods decreases.

Moreover, it should be remarked that for δ values belonging to a relative small interval around $\delta = 0$, for 3-point interpolation always the efficiency of the WMIpDFT methods is higher than the one of the IpDFT method [12].

5. CONCLUSION

This paper generalizes the results of the WMIpDFT method proposed in [1] for the case in which the maximum sidelobe decay windows with higher order than two are used in order to improve the performance in reducing the systematic errors which affect the frequency estimation of a multifrequency signal component. For this method when the H-term maximum sidelobe decay window ($H \geq 2$) is used and for ($2J + 1$) interpolation points ($1 \leq J \leq H - 1$), the analytical formula for frequency estimation is derived.

It has been shown that the proposed WMIpDFT method and the WMIpDFT method proposed in [9] have practically the same effectiveness in reducing the systematic errors and practically the same statistical efficiencies with respect to the unbiased CR lower bound. Moreover, it has been shown that the variance of δ of a multifrequency component normalized to the corresponding CR lower bound increases as the number of interpolation points increases. Thus, by increasing the number of interpolation points the systematic errors decreases, but the statistical efficiency decreases. However, from the analysis made it can be concluded that the WMIpDFT methods are well suited when the number of recorded cycles of the multifrequency component is not so larger and the signal has a relative small power noise.

The biggest advantage of the proposed WMIpDFT is that the formula used for frequency estimation is simpler than the one of the WMIpDFT method proposed in [9] and so, more easy to implement. Therefore, the proposed WMIpDFT method is the one recommended to be used.

APPENDIX

Determination the expression of α_k

Using (6) after some algebra the numerator of α_k defined in (10) becomes

$$\begin{aligned}
&\sum_{i=1}^{J} C_{2J-2}^{J-i}\left[1/(i-\delta_k)\prod_{h=1}^{H-1}\left(h^2-(i-\delta_k)^2\right)-1/(i+\delta_k)\prod_{h=1}^{H-1}\left(h^2-(i+\delta_k)^2\right)\right]\\
&-\sum_{i=1}^{J-2} C_{2J-2}^{J-i-2}\left[1/(i-\delta_k)\prod_{h=1}^{H-1}\left(h^2-(i-\delta_k)^2\right)-1/(i+\delta_k)\prod_{h=1}^{H-1}\left(h^2-(i+\delta_k)^2\right)\right]\\
&=\frac{1}{\delta_k\prod_{h=1}^{H-1}\left(h^2-\delta_k^2\right)}\left\{\sum_{i=1}^{J} C_{2J-2}^{J-i}\left[\frac{(H-i+\delta_k)(H-i+1+\delta_k)\ldots(H-1+\delta_k)}{(H-\delta_k)(H+1-\delta_k)\ldots(H+i-1-\delta_k)}\right.\right.\\
&\left.-\frac{(H-i-\delta_k)(H-i+1-\delta_k)\ldots(H-1-\delta_k)}{(H+\delta_k)(H+1+\delta_k)\ldots(H+i-1+\delta_k)}\right]\\
&-\sum_{i=1}^{J-2} C_{2J-2}^{J-i-2}\left[\frac{(H-i+\delta_k)(H-i+1+\delta_k)\ldots(H-1+\delta_k)}{(H-\delta_k)(H+1-\delta_k)\ldots(H+i-1-\delta_k)}\right.\\
&\left.\left.-\frac{(H-i-\delta_k)(H-i+1-\delta_k)\ldots(H-1-\delta_k)}{(H+\delta_k)(H+1+\delta_k)\ldots(H+i-1+\delta_k)}\right]\right\}.
\end{aligned} \tag{A.1}$$

To simplify (A.1), we demonstrate that

$$\begin{aligned}
&\sum_{i=1}^{J} C_{2J-2}^{J-i}\left[\frac{(H-i+x)(H-i+1+x)\ldots(H-1+x)}{(H-x)(H+1-x)\ldots(H+i-1-x)}\right.\\
&\left.-\frac{(H-i-x)(H-i+1-x)\ldots(H-1-x)}{(H+x)(H+1+x)\ldots(H+i-1+x)}\right]\\
&-\sum_{i=1}^{J-2} C_{2J-2}^{J-i-2}\left[\frac{(H-i+x)(H-i+1+x)\ldots(H-1+x)}{(H-x)(H+1-x)\ldots(H+i-1-x)}\right.\\
&\left.-\frac{(H-i-x)(H-i+1-x)\ldots(H-1-x)}{(H+x)(H+1+x)\ldots(H+i-1+x)}\right]\\
&=\frac{2x(2H+2J-3)!}{(2H-2)!\prod_{h=H}^{H+J-1}\left(h^2-x^2\right)},
\end{aligned} \tag{A.2}$$

in which x is a real number.

To prove (A.2), each side is multiplied by $H+i\pm x, i=0,1,\ldots,J-1,$ and then to x the values $\pm(H+i)$ are given. Thus, we obtain

$$C_{2H-2}^{0}C_{2J-2}^{J-i-1}+C_{2H-2}^{1}C_{2J-2}^{J-i-2}+\ldots+C_{2H-2}^{J-i-1}C_{2J-2}^{0}-$$
$$-\left(C_{2H-2}^{0}C_{2J-2}^{J-i-3}+C_{2H-2}^{1}C_{2J-2}^{J-i-4}+\ldots+C_{2H-2}^{J-i-3}C_{2J-2}^{0}\right) \quad (A.3)$$
$$=\frac{2(H+i)(2H+2J-3)!}{(2H+J+i-1)!(J-i-1)!}.$$

We have

$$C_{2H-2}^{0}C_{2J-2}^{J-i-1}+C_{2H-2}^{1}C_{2J-2}^{J-i-2}+\ldots+C_{2H-2}^{J-i-1}C_{2J-2}^{0} \quad (A.4)$$
$$=C_{2H+2J-4}^{J-i-1}$$

$$C_{2H-2}^{0}C_{2J-2}^{J-i-3}+C_{2H-2}^{1}C_{2J-2}^{J-i-4}+\ldots+C_{2H-2}^{J-i-3}C_{2J-2}^{0} \quad (A.5)$$
$$=C_{2H+2J-4}^{J-i-3}.$$

The equality (A.4) is true since the left side is the coefficient of x^{J-i-1} from the product of the binomials $(1+x)^{2H-2}(1+x)^{2J-2}$ and the right side is the coefficient of x^{J-i-1} from the binomial $(1+x)^{2H+2J-4}$.

The equality (A.5) is also true since the left side is the coefficient of x^{J-i-3} from the product of the binomials $(1+x)^{2H-2}(1+x)^{2J-2}$ and the right side is the coefficient of x^{J-i-3} from the binomial $(1+x)^{2H+2J-4}$.

We have the following equality

$$C_{2H+2J-4}^{J-i-1}-C_{2H+2J-4}^{J-i-3}=\frac{2(H+i)(2H+2J-3)!}{(2H+J+i-1)!(J-i-1)!}. \quad (A.6)$$

From (A.4) – (A.6) it follows that (A.3) is true and so, also (A.2) is true. By replacing $x = \delta_k$ in (A.2), the numerator of α_k becomes

$$\sum_{i=1}^{J}C_{2J-2}^{J-i}\left[1/(i-\delta_k)\prod_{h=1}^{H-1}\left(h^2-(i-\delta_k)^2\right)-1/(i+\delta_k)\prod_{h=1}^{H-1}\left(h^2-(i+\delta_k)^2\right)\right]$$
$$-\sum_{i=1}^{J-2}C_{2J-2}^{J-i-2}\left[1/(i-\delta_k)\prod_{h=1}^{H-1}\left(h^2-(i-\delta_k)^2\right)-1/(i+\delta_k)\prod_{h=1}^{H-1}\left(h^2-(i+\delta_k)^2\right)\right]$$
$$=\frac{2(2H+2J-3)!}{(2H-2)!\prod_{h=1}^{H+J-1}\left(h^2-\delta_k^2\right)}.$$
(A.7)

The expression of the denominator of α_k is [13]

$$\sum_{i=1}^{J}C_{2J}^{J-i}\left[1/(i-\delta_k)\prod_{h=1}^{H-1}\left(h^2-(i-\delta_k)^2\right)\right.$$
$$\left.+1/(i+\delta_k)\prod_{h=1}^{H-1}\left(h^2-(i+\delta_k)^2\right)\right] \quad (A.8)$$
$$+C_{2J}^{J}(1/\delta_k)\prod_{h=1}^{H-1}\left(h^2-\delta_k^2\right)=\frac{(2H+2J-2)!}{(2H-2)!\delta_k\prod_{h=1}^{H+J-1}\left(h^2-\delta_k^2\right)}.$$

From (A.7) and (A.8) it can be established

$$\alpha_k \cong \frac{\delta_k}{H+J-1}. \quad (A.9)$$

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